

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Perform FFT and IFFT on discrete time signal.	
Experiment :- 5	Date: 26/09/2026	Enrolment No:92400133050

Aim : Perform FFT and IFFT on discrete time signal.

Code :

```
import numpy as np

import matplotlib.pyplot as plt


signal = np.array([1,2,3,4,5,6,7,8])


# FFT

fft = np.fft.fft(signal)


# Real and Imaginary

real = np.real(fft)

imag = np.imag(fft)


# Magnitude and Phase

magnitude = np.abs(fft)

phase = np.angle(fft)


# Reconstructed Signal

reconstructed = np.real(np.fft.ifft(fft))


# 4 Plots

plt.figure(figsize=(10,12))


plt.subplot(4, 1, 1)

plt.stem(signal)

plt.title("Original Signal")
```

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```

plt.xlabel("Sample")
plt.ylabel("Amplitude")
for i, value in enumerate(signal):
    plt.annotate(
        f"{value}",
        (i, value),
        xytext=(0, 8),
        textcoords="offset points",
        ha="center"
    )
plt.subplot(4, 1, 2)
plt.stem(
    real,
    linefmt='C0-',
    markerfmt='C0o',
    basefmt=' ',
    label="Real Part"
)
plt.stem(
    imag,
    linefmt='C1-',
    markerfmt='C1s',
    basefmt=' ',
    label="Imaginary Part"
)
for i, value in enumerate(real):
    offset = 8 if value >= 0 else -20
    plt.annotate(

```

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```

        f"R={value:.2f}",
        (i, value),
        xytext=(0, offset),
        textcoords="offset points",
        ha="center",
        fontsize=8
    )
for i, value in enumerate(imag):
    if abs(value) < 1e-10:
        continue
    offset = 22 if value >= 0 else -32
    plt.annotate(
        f"I={value:.2f}",
        (i, value),
        xytext=(0, offset),
        textcoords="offset points",
        ha="center",
        fontsize=8
    )
plt.axhline(0, linewidth=0.8)
plt.title("FFT Result - Real and Imaginary Parts")
plt.xlabel("Frequency Index")
plt.ylabel("Value")
plt.xticks(range(len(signal)))
plt.legend()
plt.grid(True, alpha=0.3)
plt.subplot(4, 1, 3)
plt.stem(magnitude)

```

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```

plt.title("Magnitude Spectrum")

plt.xlabel("Frequency")

plt.ylabel("Magnitude")

for i, value in enumerate(magnitude):

    plt.annotate(

        f"{value:.2f}",

        (i, value),

        xytext=(0, 8),

        textcoords="offset points",

        ha="center"

    )

plt.subplot(4, 1, 4)

plt.stem(reconstructed)

plt.title("Reconstructed Signal using IFFT")

plt.xlabel("Sample")

plt.ylabel("Amplitude")

for i, value in enumerate(reconstructed):

    plt.annotate(

        f"{value:.2f}",

        (i, value),

        xytext=(0, 8),

        textcoords="offset points",

        ha="center"

    )

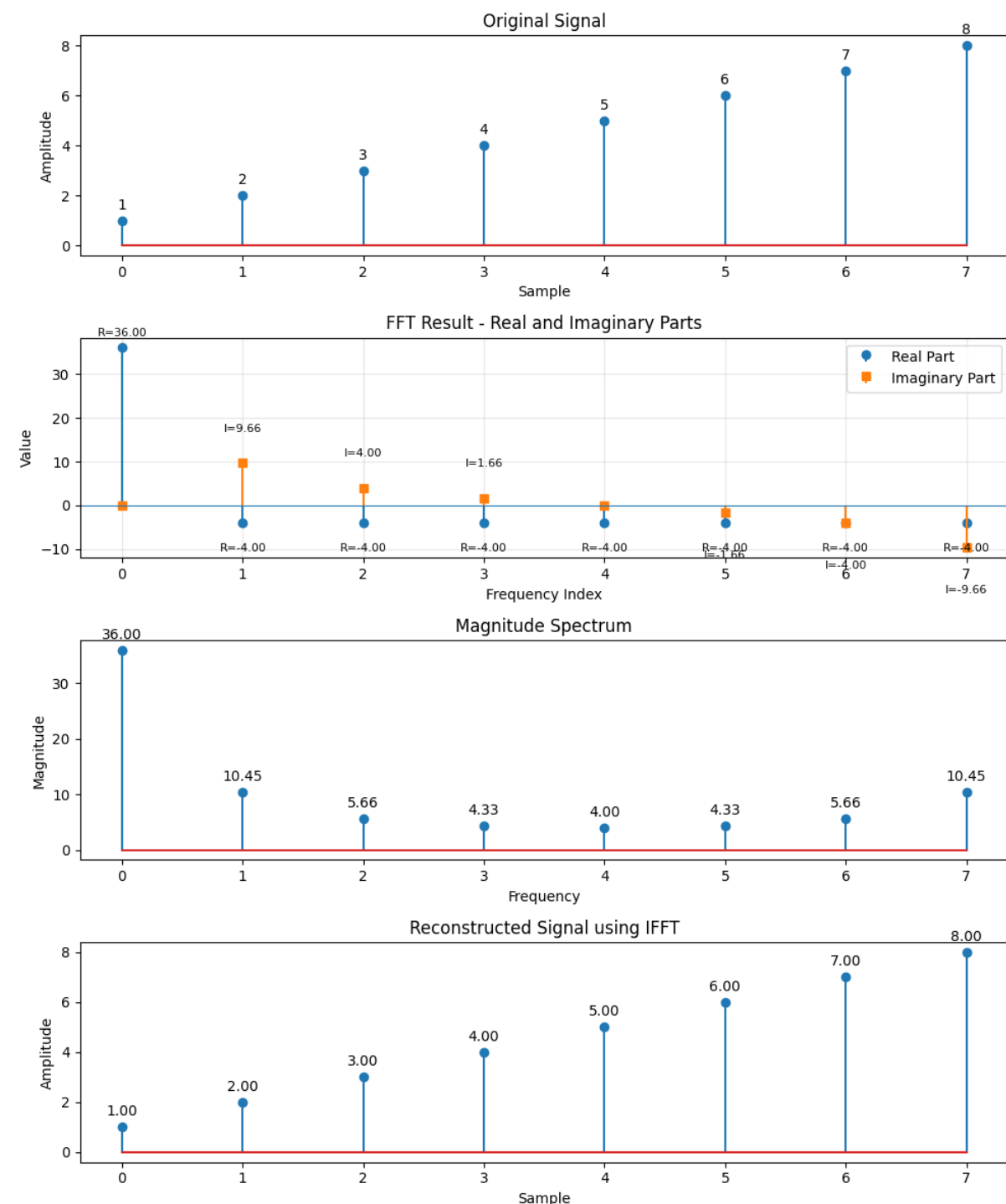
plt.tight_layout()

plt.show()

```

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Output :



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Conclusion :

In this experiment, we used Python to study a discrete-time signal in both the time domain and frequency domain. The FFT was used to break the signal into its frequency components, showing the real part, imaginary part, and magnitude spectrum. Then the IFFT was used to convert the signal back from the frequency domain to the time domain. The reconstructed signal matched the original signal. This shows that FFT is useful for analyzing a signal's frequency content, and IFFT can correctly recover the original signal from it.